

Ex. No.: 16

Date:

AIM

To implement procedures using programs in MySQL for the Hostel Management System database.

HIGH LEVEL PROGRAMMING EXTENSIONS - PROCEDURES

PROCEDURES

A procedure is a subprogram that performs a specific action. In MySQL, a stored procedure is created using CREATE PROCEDURE and executed using CALL.

Creating a Procedure

The CREATE PROCEDURE statement is used to create a stored procedure. The DELIMITER command is used to change the standard delimiter temporarily because statements inside a procedure end with a semicolon (;).

Calling a Stored Procedure

A stored procedure can be executed using:

```
CALL procedure_name();
```

Problem 1

Create a simple procedure to get all the records from the Hostel_Student table.

Program-1

```
USE HostelDB;
```

```
DROP PROCEDURE IF EXISTS GetAllStudents;
```

```
DELIMITER //
```

```
CREATE PROCEDURE GetAllStudents()
```

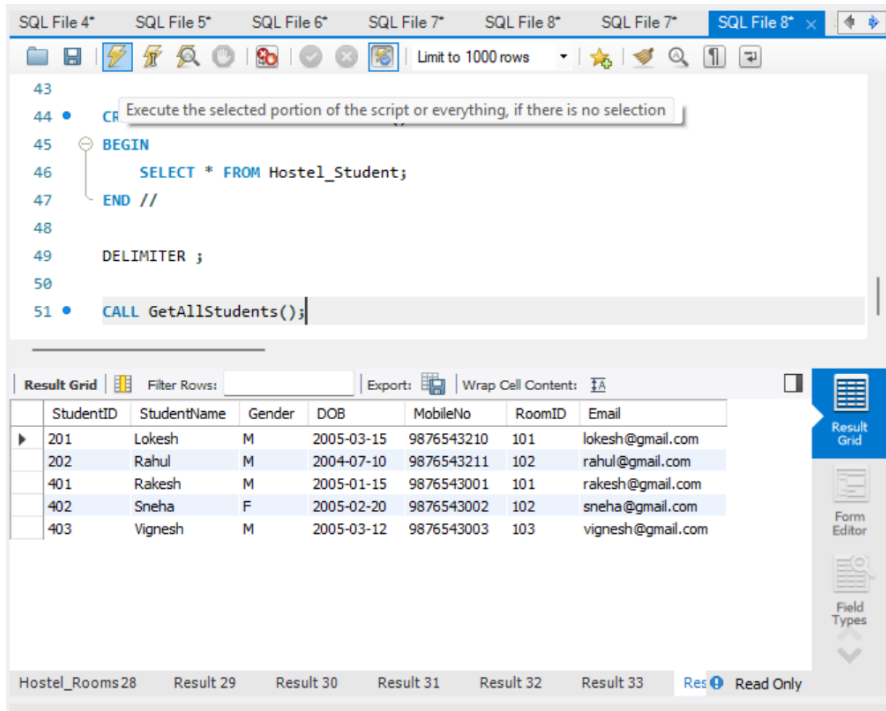
```
BEGIN
```

```
    SELECT * FROM Hostel_Student;
```

```
END //
```

```
DELIMITER ;
```

```
CALL GetAllStudents();
```



Problem 2

Create a stored procedure GetRoomLevel() that accepts a RoomID and returns the room level using RentPerMonth. If the rent is 5000 or more, the level is PLATINUM; if the rent is 4000 or more, the level is GOLD; otherwise, the level is SILVER.

Program-2

USE HostelDB;

DROP PROCEDURE IF EXISTS GetRoomLevel;

DELIMITER //

```

CREATE PROCEDURE GetRoomLevel(
    IN p_roomid INT,
    OUT p_roomlevel VARCHAR(10)
)
BEGIN
    DECLARE v_rent DECIMAL(8,2);

    SELECT RentPerMonth
    INTO v_rent
    FROM Hostel_Rooms
    WHERE RoomID = p_roomid;

```

```

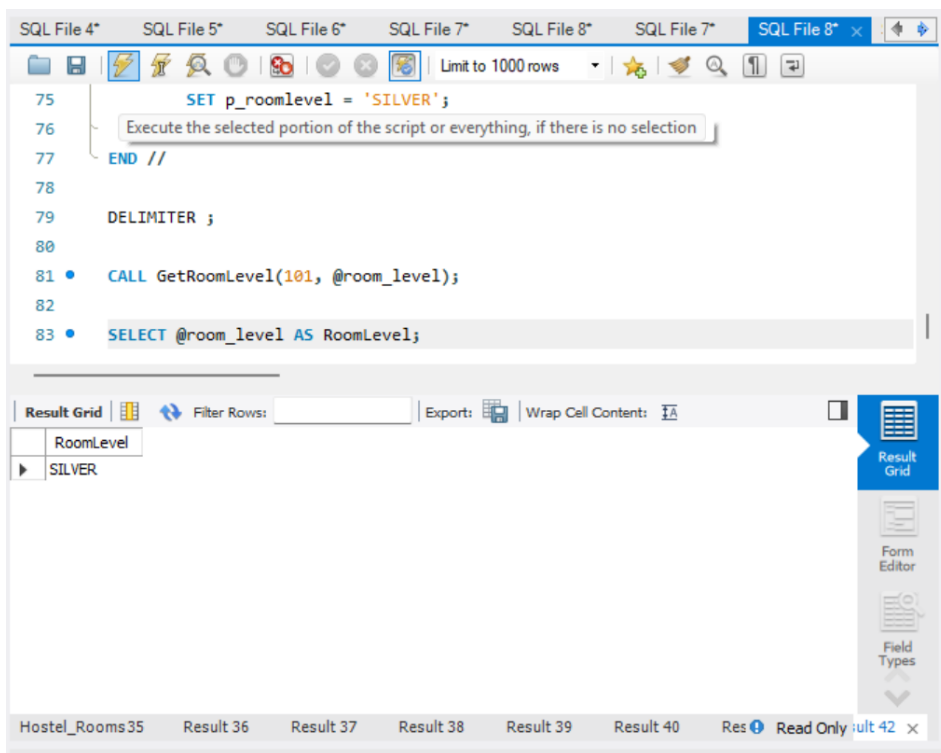
IF v_rent >= 5000 THEN
    SET p_roomlevel = 'PLATINUM';
ELSEIF v_rent >= 4000 THEN
    SET p_roomlevel = 'GOLD';
ELSE
    SET p_roomlevel = 'SILVER';
END IF;
END //

DELIMITER ;

CALL GetRoomLevel(101, @room_level);

SELECT @room_level AS RoomLevel;

```



RESULT

Thus, the high-level programming procedure programs were executed successfully using the HostelDB database.